Afternoon Naps and Word List Recall in Healthy Adults

Abstract. We tested whether a 30-minute afternoon nap improves recall of newly learned word lists.

Methods. 120 healthy adults aged 20 to 45 learned two lists of 40 words at 1 pm. Participants were randomly assigned to a nap group (n = 60), which slept for 30 minutes in a quiet room, or a wake group (n = 60), which watched a nature documentary. Recall was tested at 4 pm and again the next morning.

Results. The nap group recalled 20% more words than the wake group at 4 pm (p < 0.01). The advantage persisted the next morning at 14% (p < 0.05). The effect was larger for concrete nouns than for abstract nouns.

Limitations. Sleep was not monitored with polysomnography, so we cannot tell which sleep stages drove the effect. All participants were university staff, which limits generalization.

Conclusion. A short afternoon nap improves recall of new verbal material for at least one day.